

Fit & Well:

Core Concepts and Labs in Physical Fitness and Wellness

15th Edition
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2: Principles of Physical Fitness

Looking Ahead...

After reading this chapter, you should be able to:

- Describe how much physical activity is recommended for developing health and fitness.
- Identify the components of physical fitness and the way each component affects wellness.
- Explain the goal and basic principles of physical training.
- Describe the principles involved in designing a well-rounded exercise program.
- List the steps for making an exercise program safe, effective, and successful.

Physical Activity and Exercise for Health and Fitness

Almost any kind of physical activity promotes health.

Try to be more active during the day.

- Try to get up and move around each hour when studying, working on the computer, or watching TV.

Physical activity and exercise are points along a continuum.

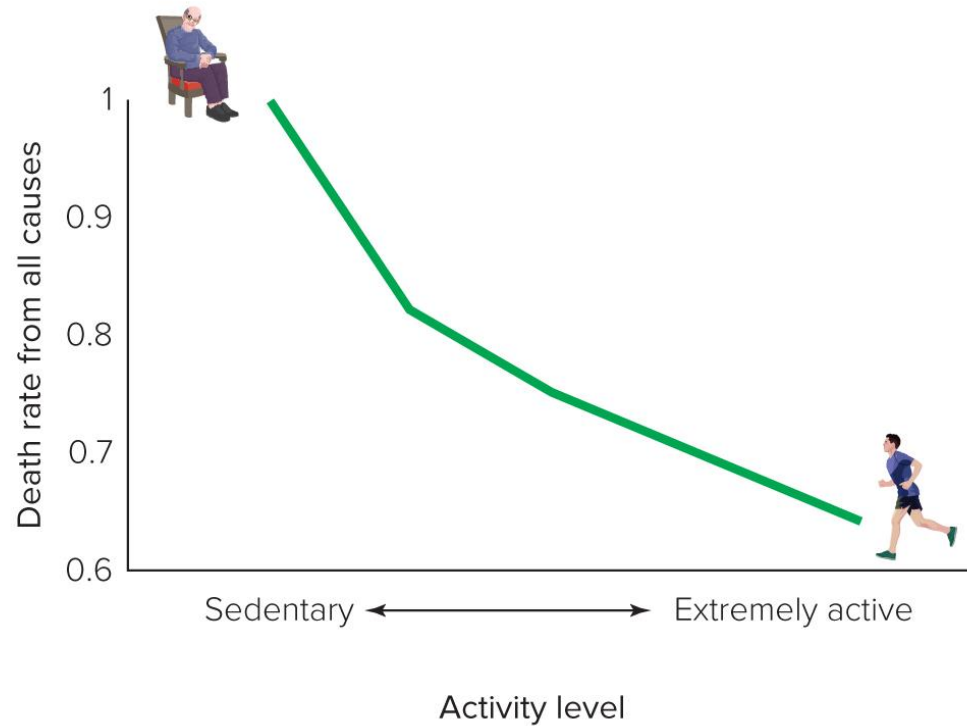


Figure 2.1 Exercise promotes longevity

The risk of death each year from all causes decreases with increased amounts and intensities of weekly physical activity.

[Access the text alternative for slide images.](#)

Source: Adapted from a composite of 12 studies involving more than 200,000 men and women. Wen, M., et al. 2014. "Physical activity and mortality among middle-aged and older adults in the United States." *Journal Physical Activity & Health* 11: 303–312.; Physical activity Guidelines Advisory Committee. *Physical Activity Guidelines Advisory Committee Report, 2008*. Washington, D.C.: U.S. Department of Health and Human Services, 2008. Schnohr, P., et al. 2015. Dose of jogging and long-term mortality: The Copenhagen City Heart Study. *Journal of the American College of Cardiology* 65(5): 411–419.

Physical Activity on a Continuum

Physical activity: body movement that is carried out by the skeletal muscles and requires energy.

Exercise: planned, structured, repetitive movement intended to improve or maintain physical fitness.

Physical activity is essential to health and confers wide-ranging health benefits, but exercise is necessary to significantly improve physical fitness.

Increasing Physical Activity to Improve Health and Wellness ¹

Recommendations for promoting health include the following:

- For substantial health benefits, adults should do at least 150 minutes of moderate-intensity or 75 minutes of vigorous-intensity aerobic physical activity each week.
- Increasing the volume and intensity of exercise results in additional health benefits.
- Adults should also do muscle-strengthening activities.
- Everyone should avoid inactivity.

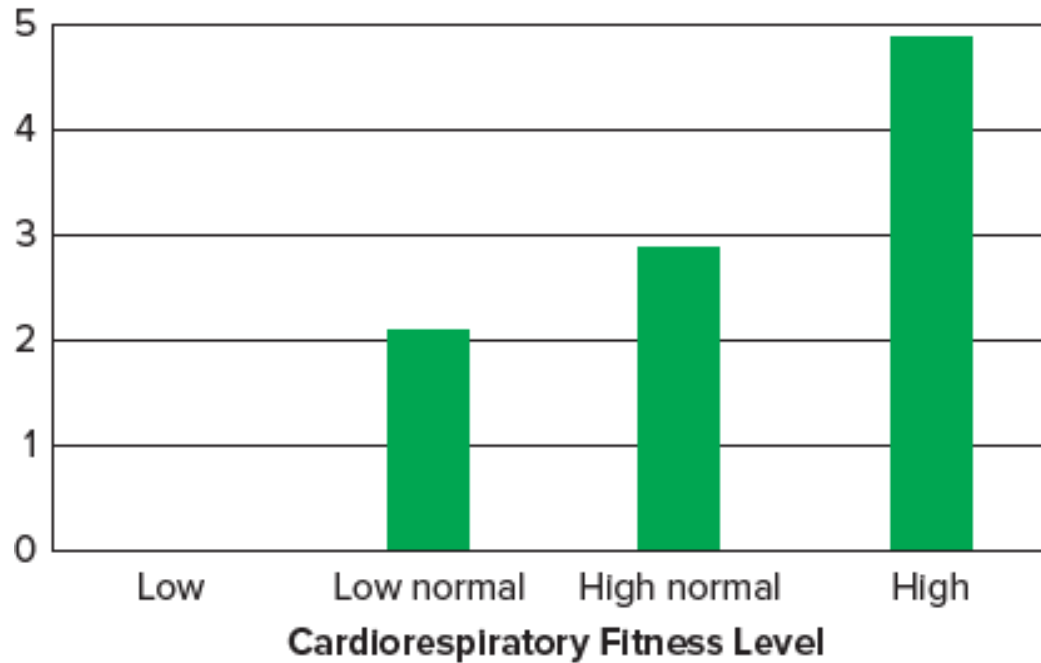


Figure 2.2 Added years of life associated with increasing fitness levels

In a study that tracked fitness levels of more than 5,000 men for 47 years, higher cardiorespiratory fitness levels were associated with increased longevity. The high fitness group lived nearly five years longer than those with the lowest fitness levels.

[Access the text alternative for slide images.](#)

Source: From the data of Clausen, Johan SR, Jacob L. Marott, Andreas Holtermann, Finn Gyntelberg, and Magnus t. Jensen. Midlife Cardiorespiratory fitness and the Long-Term risk of Mortality: 46 Years of Follow-Up. *Journal of the American College of Cardiology* 72, no. 9 (2018): 987-95.

Increasing Physical Activity to Improve Health and Wellness ₂

Physical activity benefits people of all ages and of all racial and ethnic groups, including people with disabilities.

Moderate physical activity uses about 150 calories of energy and causes a noticeable increase in heart rate.

- **Calorie:** the commonly used term for *kilocalorie* (which is actually 1,000 calories), a measure of energy equal to the heat it takes to raise the temperature of one liter of water 1°C.

Increasing Physical Activity to Improve Health and Wellness ₃

Vigorous physical activity causes rapid breathing and a substantial increase in heart rate.

The daily total of physical activity can be accumulated in multiple bouts of 10 or more minutes per day.



Figure 2.3 Examples of moderate-intensity physical activity

Each example uses about 150 calories.

Common Activities	Duration (min.)	Less Vigorous, More Time
Washing and waxing a car	45–60	
Washing windows or floors	45–60	
Gardening	30–45	
Wheeling self in wheelchair	30–40	
Pushing a stroller 1½ miles	30	
Raking leaves	30	
Walking 2 miles	30 (15 min/mile)	
Shoveling snow	15	
Stairwalking	15	
Sporting Activities		
Playing volleyball	45–60	More Vigorous, Less Time
Playing touch football	45	
Walking 1¾ miles	35 (20 min/mile)	
Basketball (shooting baskets)	30	
Bicycling 5 miles	30	
Dancing fast (social)	30	
Water aerobics	30	
Swimming laps	20	
Basketball (playing game)	15–20	
Bicycling 4 miles	15	
Jumping rope	15	
Running 1½ miles	15 (10 min/mile)	

[Access the text alternative for slide images.](#)

Table 2.1 Moderate- and Vigorous-Intensity Exercise

MODERATE-INTENSITY ACTIVITY:

Uses 3.5–7 calories per minute and causes your breathing and heart rate to increase but still allows for comfortable conversation:

- Actively playing with children or pets.
- Archery.
- Ballroom dancing.
- Bicycling or stationary bike, moderate pace.
- Downhill skiing, moderate intensity.
- Figure skating, recreational.
- Fly fishing or walking along stream.
- Gardening or yard work, moderate pace.
- Golf.
- Hiking, leisurely pace.
- Horseback riding, recreational.
- Housework, moderate intensity.
- Skateboarding.
- Softball.
- Using stair-climber, elliptical trainer, or rowing machine, moderate pace.
- Table tennis.
- Tennis, doubles.
- Walking at a moderate pace: walking to school or work, walking for pleasure.
- Water aerobics.
- Waxing the car.
- Weight training and bodybuilding.
- Yoga.

VIGOROUS-INTENSITY ACTIVITY:

Uses more than 7 calories per minute and increases your heart and breathing rates considerably. These exercises cause larger increases in physical fitness:

- Group exercise: high-impact step aerobics, aerobic dance.
- Backpacking.
- Basketball, recreational.
- Bicycling, high intensity.
- Calisthenics, vigorous: jumping jacks, burpees, air squats.
- Circuit weight training.
- Cross-country skiing or snowshoeing.
- Cross-training, such as CrossFit.
- Downhill skiing, vigorous intensity.
- Football, recreational.
- Gardening or yard work, shoveling heavy snow, digging ditches.
- Hand cycling.
- High-intensity interval training: running, elliptical trainer, swimming, cycling.
- Horseback riding, galloping or jumping.
- In line skating.
- Jogging.
- Kayaking, whitewater.
- Pushing a car.
- Running up stairs.
- Soccer, recreational.
- Tennis, singles.
- Wheelchair wheeling training.

Source: Centers for Disease Control and Prevention. 2015. *General Physical Activities Defined by Level of Intensity* (www.cdc.gov/nccdphp/dnpa/physical/pdf/PA_Intensity_table_2_1.pdf).

Increasing Physical Activity to Manage Weight

Two-thirds of Americans are overweight.

The U.S. Department of Health and Human Services has published guidelines focusing on weight management:

- To prevent weight gain, lose weight, or maintain weight-loss, 150–300 minutes per week of physical activity may not be enough.
- Up to 90 minutes of physical activity per day is recommended.

Exercising to Improve Physical Fitness

People can obtain greater health and wellness benefits by increasing the duration and intensity of physical activity.

Physical fitness requires more intense movement that poses a substantially greater challenge to the body.



Table 2.2 Physical Activity and Exercise Recommendations for Promoting General Health, Fitness, and Weight Management

GOAL	RECOMMENDATION
General health	Perform moderate-intensity aerobic physical activity for at least 150 minutes per week (30 minutes 5 times per week) or 75 minutes of vigorous-intensity physical activity per week (25 minutes 3 times per week). Also, be more active in your daily life: Walk instead of driving, take the stairs instead of the elevator, and watch less TV.
Increased health benefits	Exercise at moderate intensity for 300 minutes per week or at vigorous intensity for 150 minutes per week.
Achieve or maintain weight loss	Exercise moderately for 60–90 minutes per day on most days of the week.
Muscle strength and endurance	Perform one or more sets of resistance exercises that work the major muscle groups for 8–12 repetitions (10–15 reps for older adults) on at least two non-consecutive days per week. Examples include weight training and exercises that use body weight as resistance (such as core-stabilizing exercises, pull-ups, push-ups, lunges, and squats).
Flexibility	Perform range-of-motion (stretching) exercises at least two or three days per week. Hold each stretch for 10–30 seconds.
Neuromuscular training	Older adults should do balance training at least two or three days per week. Examples include yoga, tai chi, and balance exercises (standing on one foot, step-ups, and walking lunges). These exercises are beneficial for young and middle-aged adults, as well.

Sources: American College of Sports Medicine. 2022. *ACSM's Guidelines for Exercise Testing and Prescription*, 11th ed. Philadelphia: Wolters Kluwer. Garber, C.E., et al. 2011. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: Guidance for prescribing exercise. *Medicine & Science in Sports & Exercise* 43(7): 1334–1359; 2018 Physical Activity Guidelines Advisory Committee. 2018. *2018 Physical Activity Guidelines Advisory Committee Scientific Report*. Washington, DC: U.S. Department of Health and Human Services.

How Much Physical Activity Is Enough?

Regular physical activity, regardless of intensity, makes you healthier and can help protect you from chronic disease.

You obtain even more benefits when you are physically fit.

- Fit people have more energy and better body control and can enjoy a more active lifestyle.

Some physical activity is better than *none*.

More physical activity is better than *some*—as long as it does not result in injury.

Components of Physical Fitness

Health-related fitness—physical capacities that contribute to health—includes the following components:

- Cardiorespiratory endurance.
- Muscular strength.
- Muscular endurance.
- Flexibility.
- Body composition.

Cardiorespiratory Endurance

Cardiorespiratory endurance: the ability of the body to perform prolonged, large-muscle, dynamic exercise at moderate to high levels of intensity.

- It depends on such factors as the ability of the lungs to deliver oxygen to the bloodstream.
- **Oxygen:** an element that is critical for generating usable energy in the body and is an important component of carbohydrates, fats, and proteins.

Cardiorespiratory endurance is a central component of health-related fitness because heart and lung function is so essential to overall good health.

Muscular Strength

Muscular strength: the ability of a muscle to exert force in a single maximum effort.

- **Relative strength:** the maximum force exerted, relative to body weight, body size, and muscle size.
- Greater muscle mass means a higher rate of **metabolism**—the sum of all the vital processes by which food energy and nutrients are made available to and used by the body.

Maintaining strength and muscle mass is vital for healthy aging.

Muscular Endurance

Muscular endurance: the ability of a muscle to remain contracted or to contract repeatedly for a long period of time.

Muscular endurance is important for good posture and for injury prevention.

Flexibility

Flexibility: the ability to move joints through their full ranges of motion.

- It depends on joint structure, the length and elasticity of connective tissue, and nervous system activity.

Inactivity causes the joints to become stiffer with age.

- That stiffness often leads people to assume unnatural postures that can stress joints and muscles.
- Stretching can help ensure a healthy range of motion.

Body Composition

Body composition: the proportion of fat and fat-free mass (muscle, bone, and water) in the body.

- Healthy body composition involves a high proportion of fat-free mass and an acceptably low level of body fat.
- **Fat-free mass:** the nonfat component of the human body, consisting of skeletal muscle, bone, and water.

Somatotype affects a person's choice of exercise.

- **Somatotype:** a body-type classification system that describes people as predominantly muscular (mesomorph), tall and thin (ectomorph), or round and heavy (endomorph).

Skill (Neuromuscular)–Related Components of Fitness

Skill (neuromuscular)–related fitness: physical capacities that contribute to performance in a sport or an activity.

- These include speed, power, agility, balance, coordination, and reaction and movement time.

Skill-related fitness tends to be sport-specific and is best developed through practice.



Principles of Physical Training: Adaptation to Stress

The human body is adaptable, adjusting to meet increasing demands.

- **Adaptation:** the physiological changes that occur with exercise training.

Short-term adjustments lead to long-term changes and improvements in fitness levels.

Physical training: the performance of different types of activities that cause the body to adapt and improve its level of fitness.

Specificity—Adapting to Type of Training

Specificity: the training principle that developing a particular fitness component requires performing exercises specifically designed for that component.

A well-rounded exercise program should include exercises geared to each component of fitness.

Progressive Overload— Adapting to the Amount of Training

Progressive overload: the training principle that progressively increasing amounts of stress on the body causes adaptation that improves fitness.

The amount of new activity added above a person's usual level is known as *overload*.

The amount of overload is important.

- The exercise needed to improve fitness depends on the individual.

The FITT Principle

The overload needed to maintain or improve a particular level of fitness for a particular fitness component is determined through four dimensions known as *FITT*:

- Frequency—how often.
- Intensity—how hard or how fast.
- Time—how long (duration).
- Type—mode of activity.

Additional components to consider:

- Volume—how much (frequency $\times$ intensity $\times$ time).
- Progression—how a program advances over time.

Reversibility—Adapting to a Reduction in Training

Fitness is a reversible adaptation.

- The body adjusts to lower levels of physical activity the same way it adjusts to higher levels.

Reversibility: the training principle that fitness improvements are lost when demands on the body are lowered.

If a person stops exercising, up to 50% of fitness improvements are lost within two months.

Individual Differences— Limits on Adaptability

There are large differences in our ability to improve fitness, achieve a desirable body composition, and learn and perform sports skills.

- People respond to training at different rates.
- Specific genes influence body fat, strength, and endurance.

However, physical training improves fitness regardless of heredity.



Designing Your Own Exercise Program ¹

Getting medical clearance:

- Risk from death in exercise is small.
- People of any age who are not at high risk for serious health problems can safely exercise at a moderate intensity.
- The Physical Activity Readiness Questionnaire for Everyone (PAR-Q+) can help evaluate exercise safety.
- Physicians may recommend an **exercise stress test** or a **graded exercise test (GXT)** to determine if a person might have heart problems when exercising.

Designing Your Own Exercise Program ₂

Assessing yourself:

- Assess your current level of fitness for each of the five health-related fitness components.

Setting goals:

- The ultimate goal of every health-related fitness program is wellness that lasts a lifetime.
- Your specific goals must be important enough to you to keep you motivated.
- First, think carefully about your goals, and be clear about why you are starting a program.

Designing Your Own Exercise Program ³

Choosing activities for a balanced program:

- An ideal fitness program combines a physically active lifestyle with systematic exercise to maintain physical fitness.
- Cardiorespiratory endurance is developed by continuous rhythmic movements of large-muscle groups.
- Muscular strength and endurance can be developed through resistance training or calisthenics.
- Flexibility is developed by stretching the major muscle groups regularly and with proper technique.
- Healthy body composition can be developed through a sensible diet and regular exercise.

Sedentary Activities

Watching television, surfing the internet, talking on a landline.



Limit Your Sedentary Activities

Strength Training

2–3 nonconsecutive days per week
(all major muscle groups)

Biceps curls, push-ups, abdominal curls, bench press, calf raises



Flexibility Training

At least 2–3 days per week, ideally
5–7 days per week (all major joints)

Calf stretch, side lunge, step stretch, hurdler stretch



Cardiorespiratory Endurance Exercise

3–5 days per week (20–60 minutes
per day)



Walking, jogging, bicycling, swimming,
aerobic dancing, in-line skating,
cross-country skiing, dancing,
basketball

Moderate-Intensity Physical Activity
150–300 minutes per week; for
weight loss or prevention of weight
regain following weight loss,
60–90 minutes per day



Walking to the store or bank,
washing windows or your car,
climbing stairs, working in
your yard, walking your dog,
cleaning your room

Figure 2.4 Physical activity pyramid

[Access the text alternative for slide images.](#)

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**Table 2.3 ACSM Exercise Recommendations for Healthy Adults:
Develop and Maintain Cardiorespiratory Endurance and Body Composition**

COMPONENT	RECOMMENDATION
Frequency of training	At least five days per week for moderate-intensity exercise and at least three days per week for vigorous-intensity exercise.
Intensity of training	55/65–90% of maximum heart rate or 40/50–85% of heart rate reserve or oxygen uptake reserve. (Reserve refers to the difference between resting and maximum values of heart rate or oxygen consumption.) The lower-intensity values (55–64% of maximum heart rate and 40–49% of heart rate reserve plus rest) are most applicable to unfit individuals. For average individuals, intensities of 70–85% of maximum heart rate or 60–80% of heart rate reserve plus rest are appropriate. Well-trained people can train at near maximum intensities. These methods increase exercise intensity within the limits of each person’s reserve capacity.
Time (duration) of training	20–60 total minutes per day of continuous or intermittent (in sessions lasting 10 or more minutes) aerobic activity. Duration depends on the intensity of activity; thus, low-intensity activity should be conducted over a longer period of time (30 minutes or more). Low- to moderate-intensity activity of longer duration is recommended for nonathletic adults.
Type (mode) of activity	Any activity that uses large-muscle groups, can be maintained continuously, and is rhythmic and aerobic in nature—for example, walking-hiking, running-jogging, bicycling, cross-country skiing, aerobic dancing and other forms of group exercise, rope-skipping, rowing, stair-climbing, swimming, skating, and endurance game activities.
Volume	Expend the equivalent of at least 1,000 calories per week, which is equivalent to about 150 minutes per week of moderate-intensity exercise.
Progression	Adjust frequency, intensity, and/or time until you reach your goal.

Chapter 3 has instructions for calculating target heart rate intensity for cardiorespiratory endurance exercise.

Sources: American College of Sports Medicine. 2022. ACSM’s Guidelines for Exercise Testing and Prescription, 11th ed. Philadelphia: Wolters Kluwer; Garber, C. E., et al. 2011. “Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: Guidance for prescribing exercise,” *Medicine and Science in Sports and Exercise* 43(7): 1334–1359.

Table 2.3 ACSM Exercise Recommendations for Healthy Adults: Develop and Maintain Muscular Strength and Endurance, Flexibility, and Body Composition

COMPONENT	RECOMMENDATION
Resistance training	One set of 8–10 exercises that condition the major muscle groups, performed two to three days per week. Most people should complete 8–12 repetitions of each exercise to the point of fatigue; practicing other repetition ranges (for example., 3–5 or 12–15) also builds strength and endurance; for older and frailer people (approximately 50–60 and older), 10–15 repetitions with a lighter weight may be more appropriate. Multiple-set regimens will provide greater benefits if time allows. Any mode of exercise that is comfortable throughout the full range of motion is appropriate (for example., free weights, kettlebells, calisthenics, elastic bands, or weight machines).
Flexibility training	Static stretches, performed for the major muscle groups at least two to three days per week, ideally daily. Stretch to the point of tightness, holding each stretch for 10–30 seconds; perform 2–4 repetitions of each stretch, for a total of 90 seconds of stretching time for each exercise.

Sources: American College of Sports Medicine. 2022. *ACSM's Guidelines for Exercise Testing and Prescription*, 11th ed. Philadelphia: Wolters Kluwer; Garber, C. E., et al. 2011. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: Guidance for prescribing exercise, *Medicine and Science in Sports and Exercise* 43(7): 1334–1359.

	 Lifestyle physical activity	 Moderate exercise program	 Vigorous exercise program
Description	Moderate physical activity (150 minutes per week; muscle-strengthening exercises 2 or more days per week)	Cardiorespiratory endurance exercise (20–60 minutes, 3–5 days per week); strength training (2–3 nonconsecutive days per week); and stretching exercises (2 or more days per week)	Cardiorespiratory endurance exercise (20–60 minutes, 3–5 days per week); interval training; strength training (3–4 nonconsecutive days per week); and stretching exercises (5–7 days per week)
Sample activities or program	• Walking to and from work, 15 minutes each way • Cycling to and from class, 10 minutes each way • Doing yard work for 30 minutes • Dancing (fast) for 30 minutes • Playing basketball for 20 minutes • Muscle exercises such as push-ups, squats, or back exercises	• Jogging for 30 minutes, 3 days per week • Weight training, 1 set of 8 exercises, 2 days per week • Stretching exercises, 3 days per week	• Running for 45 minutes, 3 days per week • Intervals, running 400 m at high effort, 4 sets, 2 days per week • Weight training, 3 sets of 10 exercises, 3 days per week • Stretching exercises, 6 days per week
Health and fitness benefits	Better blood cholesterol levels, reduced body fat, better control of blood pressure, improved metabolic health, and enhanced glucose metabolism; improved quality of life; reduced risk of some chronic diseases Greater amounts of activity can help prevent weight gain and promote weight loss	All the benefits of lifestyle physical activity, plus improved physical fitness (increased cardiorespiratory endurance, muscular strength and endurance, and flexibility) and even greater improvements in health and quality of life and reductions in chronic disease risk	All the benefits of lifestyle physical activity and a moderate exercise program, with greater increases in fitness and somewhat greater reductions in chronic disease risk Participating in a vigorous exercise program may increase risk of injury and overtraining

Figure 2.5 Health and fitness benefits of different amounts of physical activity and exercise

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Designing Your Own Exercise Program ⁴

Guidelines for training:

Train the way you want your body to change.

Train regularly.

Start slowly, and get in shape gradually.

- *Beginning phase*: the body adjusts.
- *Progress phase*: fitness increases.
- *Maintenance phase*: the targeted level of fitness is sustained over the long term.
- Increase duration and frequency before intensity.
- **Overtraining**: a condition caused by training too much or too intensely, resulting in lack of energy, decreased physical performance, and aching muscles and joints.

Designing Your Own Exercise Program ⁵

Guidelines for training, continued:

Warm-up before exercise.

Cool down after exercise.

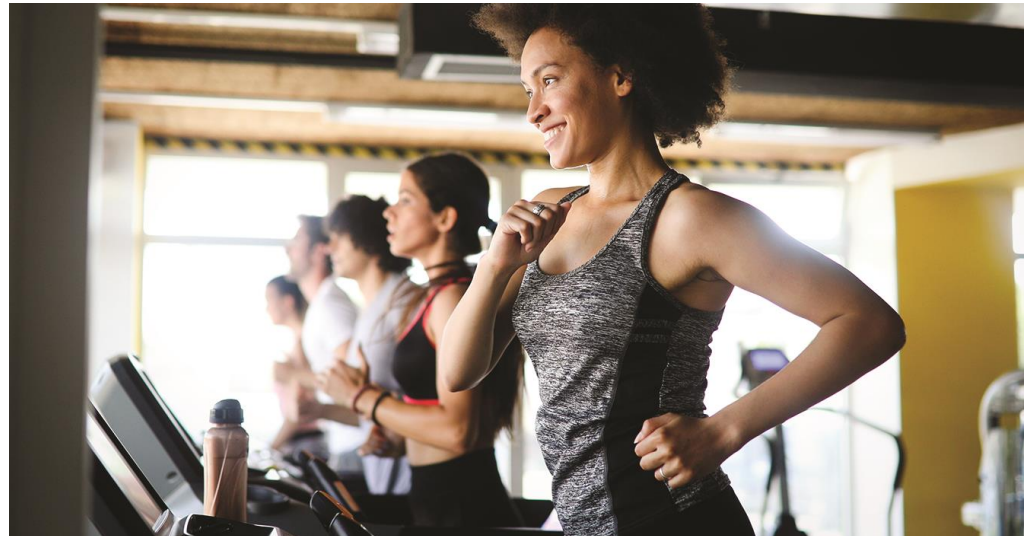
Exercise safely.

- Protect yourself with the appropriate clothing and gear.
- Take a partner when exercising in deserted areas.
- Give vehicles plenty of leeway.
- Take precautions in the weight room.
- Always use good technique.
- Overloading your muscles and joints can lead to serious injury, so train within your capacity.

Designing Your Own Exercise Program ⁶

Guidelines for training, continued:

- Listen to your body and get adequate rest.
- Cycle the volume and intensity of your workouts.
- Vary your activities.
- Train with a partner.



Designing Your Own Exercise Program 7

Guidelines for training, continued:

Train your mind.

- Becoming fit requires commitment, discipline, and patience.

Fuel your activity appropriately.

Have fun.

Track your progress.

Get help and advice if you need it.

Keep your exercise program in perspective.

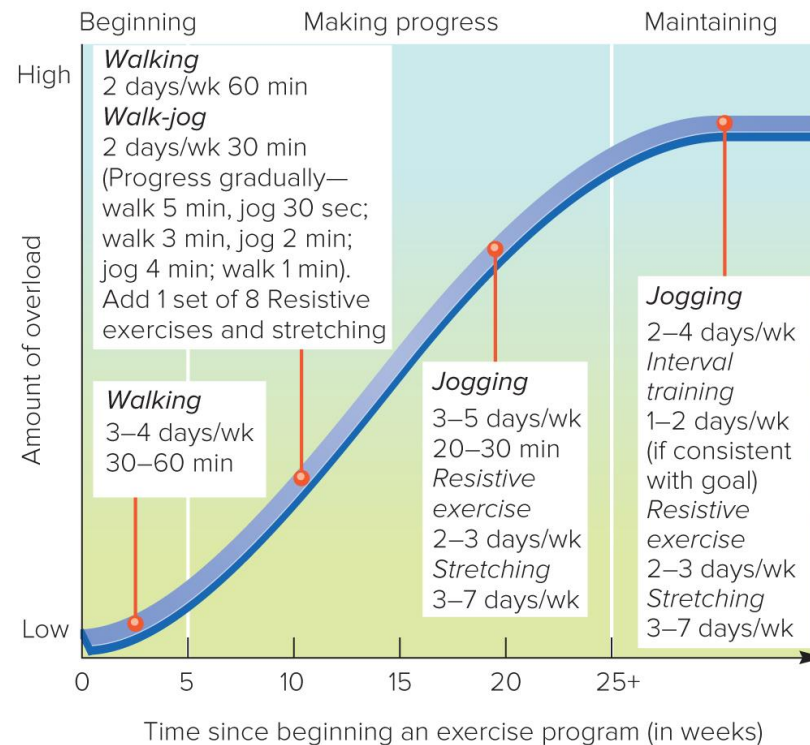


Figure 2.6 Progression of an exercise program

This figure shows how you can increase the amount of overload gradually in a walking and running program. Begin with slow walking for several weeks and gradually pick up the pace. Gradually introduce short periods of running during your walks, progressively running farther and walking less until you can run continuously. Add interval training after four to six weeks of jogging—if you want to reach higher levels of fitness. Regardless of the activity and the exercise program, begin slowly and progress gradually. After you achieve the desired level of fitness, you can maintain it by exercising three to five days a week.

[Access the text alternative for slide images.](#)

SOURCE: Progression data from American College of Sports Medicine. 2022. *ACSM's Guidelines for Exercise Testing and Prescription*, 11th ed. Philadelphia: Wolters Kluwer/Lippincott Williams & Wilkins Health.

Test Your Knowledge

Is it true that to improve your health, you must exercise vigorously for at least 30 minutes straight, five or more days per week?

Which activity uses about 150 calories?

- Washing a car for 45–60 minutes;
- Shooting a basketball for 30 minutes;
- Jumping rope for 15 minutes; or
- All three?

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